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Interim Submission

Research Paper Summary

Paper 1:

Tschoellitsch et al. (2023)

Using emergency department triage for machine learning-based admission and mortality prediction

This paper addresses the problem that emergency departments frequently are required to make admission and risk decisions using sparse early patient data. The authors, Tschoellitsch et al. (2023), used a single-centre retrospective cohort of 58,323 adult patients triaged with the Manchester Triage System [1]. Then, the authors trained machine learning models including Random Forests and Neural Networks to predict the ward observation, ICU admission and 30 day mortality. The study reports that the models performed well. Highlighting that they achieved AUC-ROC scores of 0.842 for the ward observation, 0.819 for ICU admission, and 0.925 for 30 day mortality. The important features included age, chief complaint, Manchester Triage category, heart rate, and oxygen saturation. The authors stated that the limitations included the single-centre design, unbalanced outcome groups (especially for ICU admission and 30-day mortality), missing or incomplete data, possible measurement errors. They also noted broader limitations of machine learning in medicine, including scarce clinical data, possible need for different thresholds or cost matrices and the need for external validation in larger and different populations.

Additionally, it should be noted that the AUC-ROC value utilized throughout the study is defined as the Area Under the Receiver Operating Characteristic Curve which can be defined as a robust metric that evaluates the goodness of the model across the full spectrum of thresholds. Specifically, the AUC-ROC is a single scalar value that facilitates the ease of comparing multiple ML models within the study in a way that evaluates them regardless of their classification thresholds.

Paper 2:

Agius et al. (2025)

Predicting Emergency Severity Index (ESI) level, hospital admission, and admitting ward in an emergency department using data-driven machine learning

This paper addresses the problem of improving emergency department triage decisions. Specifically, Aguis et al. (2025) focus on the patient prioritisation, admission prediction, and admitting ward prediction [2]. The authors used a large retrospective dataset of 653,546 emergency department visits from Mater Dei Hospital in Malta and developed two XGBoost (Extreme Gradient Boosting algorithm) models in 2 stages. Stage 1 used demographic and presenting symptom data available at triage, while Stage 2 added blood test results such as Haemoglobin, C-Reactive Protein and White Blood Cell Count to enhance the predictions. The reported outcomes showed improved performance after adding clinical data: prioritisation accuracy increased from 0.75 to 0.76, admission prediction accuracy from 0.80 to 0.82 and

admitting ward prediction accuracy from 0.80 to 0.86. The authors stated that the study was limited by data quality issues, possible bias from missing data handling and SMOTE oversampling, reduced XGBoost interpretability, single hospital generalisability, possible changes in healthcare practice over time, overfitting risk, and ethical concerns around using personal health data in predictive models.

Paper 3:

Wang et al. (2024)

Use of Mobile Technologies to Streamline Pretriage Patient Flow in the Emergency Department

Wang et al. (2024) addressed the problem of inefficient emergency department pretriage flow, where nurses must quickly assess patient acuity while also recording patient information [3]. The authors tested a smartphone and QR code–based self-registration system in three busy emergency departments in Shanghai and compared it with kiosk-based registration using waiting time, usability scores, nurse Net Promoter Score, and qualitative observations. The study found that smartphone self-registration reduced referring effort from 43.7% to 8.8% and reduced mean pretriage waiting time from 4.4 to 2.9 minutes, while also producing higher usability and nurse satisfaction scores. This paper is relevant to a triage system pilot because it shows that digitising patient intake can improve pretriage workflow before predictive decision-support tools are added. The authors stated that the study was limited by being conducted only in three metropolitan EDs in Shanghai, meaning the findings may not generalise to other regions or ED volumes; they also noted the short observation period and that the smartphone tool lacked future features such as mobile sensor integration and triage-prediction algorithms.

References

- [1] Tschoellitsch T, Seidl P, Böck C, Maletzky A, Moser P, Thumfart S, et al. Using emergency department triage for machine learning-based admission and mortality prediction. *European Journal of Emergency Medicine*. 2023 Aug 14;30(6):408–16. doi:10.1097/MEJ.0000000000001068.

- [2] Agius S, Cassar V, Magri C, Khan W, Al-Jumeily D, Caruana G, et al. Predicting Emergency Severity Index (ESI) level, hospital admission, and admitting ward in an emergency department using data-driven machine learning. *BMC Medical Informatics and Decision Making* 2025;25(1):281. doi:10.1186/s12911-025-02941-9.

- [3] Wang P, Yu L, Li T, Zhou L, Ma X. Use of mobile technologies to streamline pretriage patient flow in the emergency department: observational usability study. *JMIR Mhealth Uhealth*. 2024;12:e54642. doi:10.2196/54642.

Problem Statement

Emergency departments must prioritise patients quickly but early triage decisions are often made with limited data while nurses manage registration, documentation and patient flow. Existing studies show that machine learning can predict admission, ICU need and mortality, while mobile self-registration can reduce pre-triage delays. However, these approaches are often tested in single settings, depend on digitised data and may lack interpretability or local validation. This project will explore a lightweight digital triage support system that captures early patient data and supports safer, faster prioritisation.